Getamazednow AI

A Getamazednow Company

MOCK / PORTFOLIO REFERENCE

GenAI Project Observability

A reference architecture and demo dashboard for observing agentic Gen AI workflows in production — built for architecture-board, risk-committee and Community of Practice review ahead of a real Datadog implementation.

MOCK / SYNTHETIC DATA — NOT CONNECTED TO ANY LIVE SYSTEM

Executive Overview | 2 July 2026

AGENDA

What we'll cover

01

What this demo is

The problem it solves, the scenario, the headline numbers

02

How to use the dashboard

A 7-tab tour built for a live walkthrough

03

Architecture, caveats & constraints

What's real, what's mocked, and why — stated up front

04

Roadmap & next steps

The 90-day path from this demo to a live pilot

01

What this demo is

Why conventional APM misses agentic Gen AI failure modes — and what this repo shows instead

01 — THE PROBLEM

A “successful” response can still hide failure

A Gen AI agent can return an HTTP 200 and still have failed. Conventional APM misses what matters most:

Conventional APM sees	Agentic Gen AI actually needs to catch
Request succeeded (200 OK)	Was the business outcome achieved, or did the agent produce a wrong or ungrounded answer?
Latency within SLA	Was cost within budget — did the workflow quietly burn 10x the expected tokens?
No application errors	Did a prompt-injection attempt bypass a guardrail and reach a money-moving tool?
Server didn't crash	Did an agent loop fan out into 40 tool calls before anyone noticed?

This is the gap the two source documents address — cost, reliability, performance, agent behaviour, security and responsible-AI risk, correlated by trace, in one observability model.

Order Support & Returns Assistant

An agentic retail customer-service workflow — order status, returns, refunds, loyalty credits, address changes — across three storefront brands, over a synthetic 30-day window (1–30 June 2026).

- **Cost at scale —** thousands of workflows a month, real token/tool cost economics.
- **Tool-driven side effects —** refunds move money; this is not a read-only chatbot.
- **A believable attack surface —** prompt injection against a refund_issue tool is realistic and high-stakes.

TWO SOURCE DOCUMENTS

AI Observability Metrics Catalogue for Agentic LLM Workflows (v1.0)
— the what to measure and govern baseline: ~150 metrics across 13 dimensions.

AI Observability — Datadog Implementation Addendum
— the how to implement it companion: telemetry contract, 7-dashboard pack, SLOs, 90-day roadmap.

Three storylines, deliberately built in

REL-2026-0608 / 0609

Prompt v14 → v15 degraded groundedness & citation accuracy — caught by the eval harness, rolled back ~27h later.

INC-2026-0611 (Sev2)

Prompt-injection cluster targeting the refund_issue tool — guardrail caught 92%, a handful bypassed.

INC-2026-0621 (Sev1)

Promo traffic surge triggered provider rate-limiting — retries, fallback routing, cost & latency spike.

Each incident is a rehearsal for a real production scenario — the dashboards show a diagnosable signal, not noise.

From the current synthetic 30-day run

~5,100
Workflows (30 days)

~95%
Success rate

~$0.025
Cost per successful workflow

~$120 / $150
Monthly AI run-rate vs. budget

~$31,400
Est. monthly cost avoidance*

2 + 1
Incidents (1 Sev1, 1 Sev2) + rollback

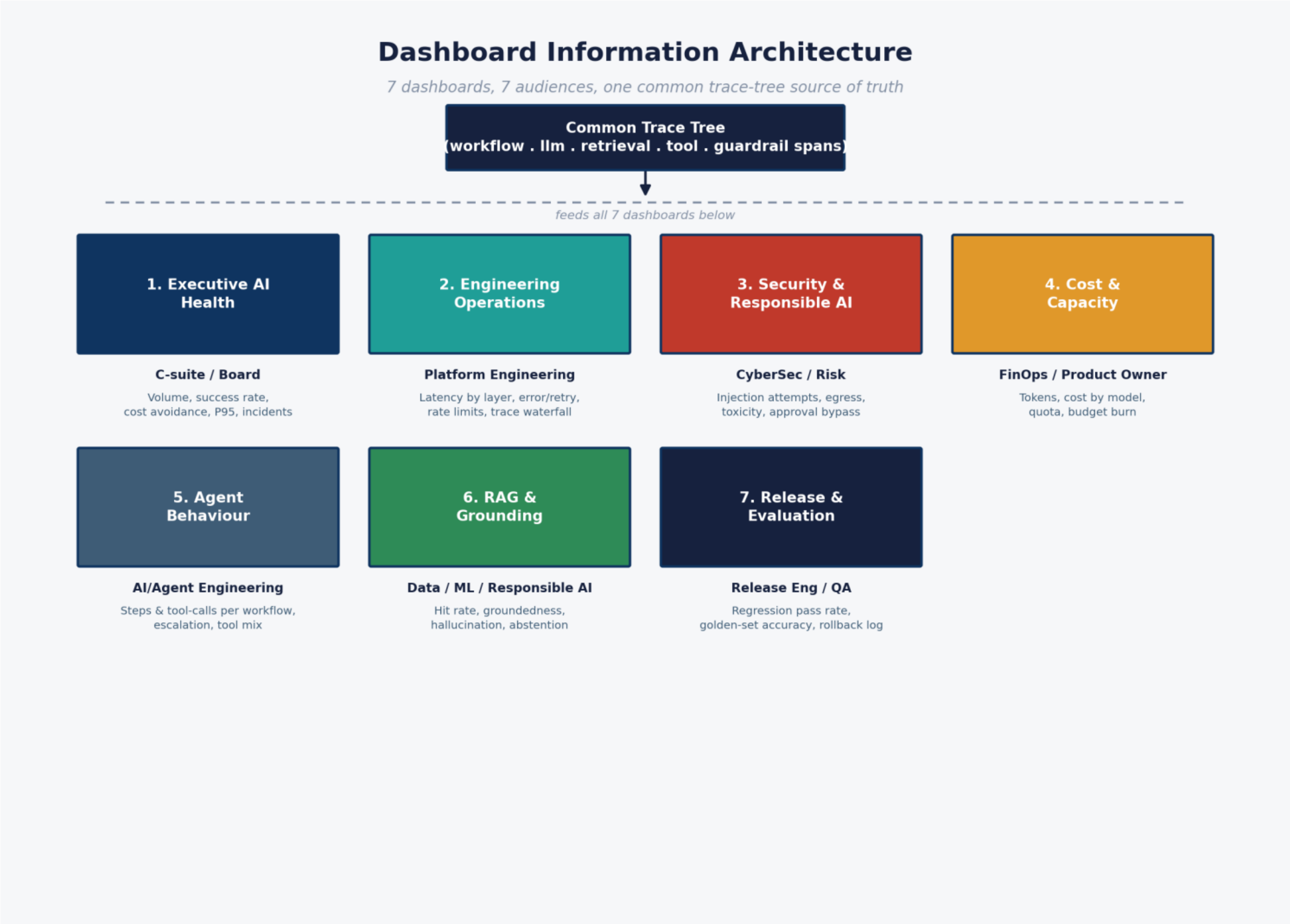
**Illustrative only — assumes a $6.50 fully-loaded cost per human-handled contact, a modelling assumption for this demo, not a sourced benchmark.*

02

How to use the dashboard

A 7-tab tour, one dashboard per audience, all reading from a single trace-tree source of truth

7 dashboards, 7 audiences, one trace tree



Suggested walkthrough order

- **1. Executive AI Health —** set business context and headline numbers.
- **2. Security & Responsible AI —** open the incident spotlight; usually the most attention-grabbing tab.
- **3. Engineering Operations —** the trace waterfall grounds the abstract “trace tree” in a concrete picture.
- **4. Cost & Capacity —** connect the rate-limit incident to a dollar figure.
- **5. RAG & Grounding or Release & Evaluation —** close by flagging the synthetic-eval-harness caveat explicitly.

RUNNING IT

```
cd genai-observability-demo/dashboard
python3 -m http.server 8000
# open http://localhost:8000
```

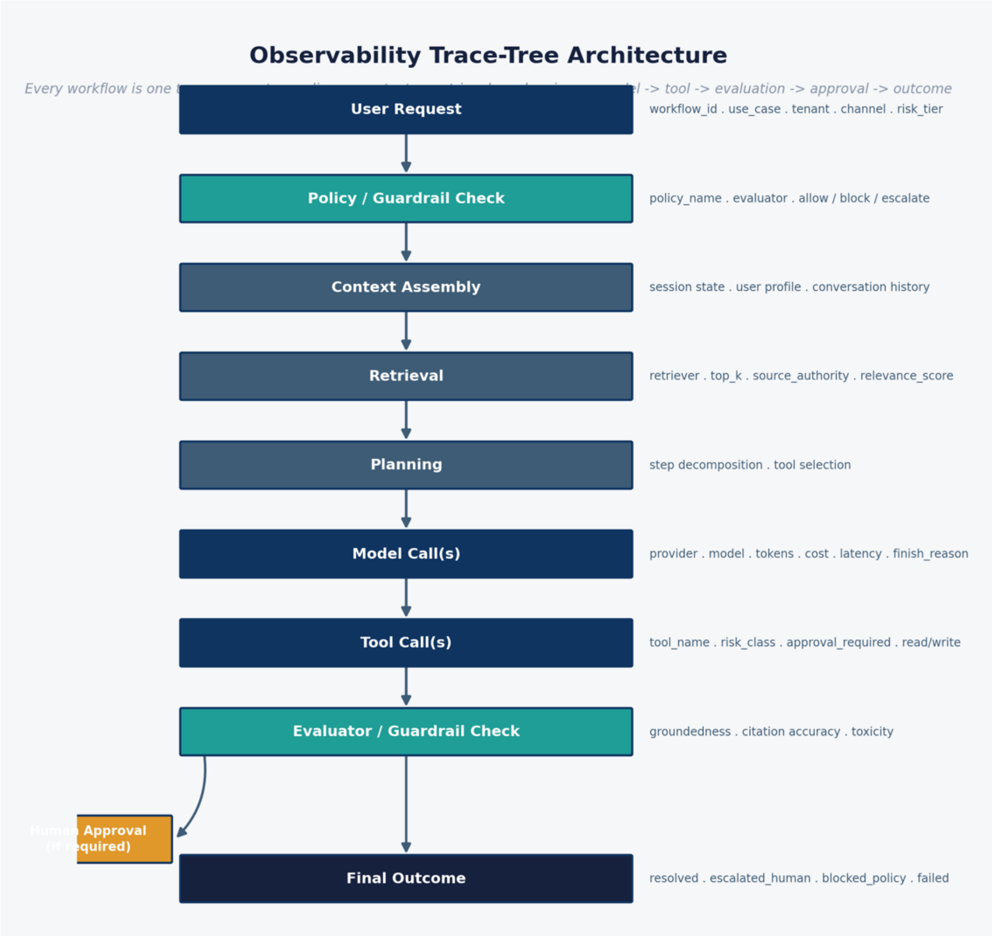
Static HTML/CSS/JS, Chart.js via CDN. No build step. Every screen carries a “MOCK / SYNTHETIC DATA” pill — keep it visible in any live demo.

03

Architecture, caveats & constraints

Honesty about scope is the point — what's real, what's mocked, and why

One connected trace tree, not isolated logs



Architectural principle: implement as ONE connected trace tree per workflow, never isolated logs.
Source: Metrics Catalogue v1.0, section 1

POSITIONING

Datadog is the production observability/control plane — not the policy authority. Policy ownership stays with architecture, security, data governance and responsible-AI risk.

The synthetic guardrail spans record policy decisions (allow/block/escalate); they do not implement policy.

This single choice — a connected trace tree — is what makes cross-dimensional correlation possible: the same workflow_id that shows up as a cost line in the Executive tab is the same trace that shows a policy bypass in the Security tab.

The single most important caveat

Synthetic evaluation-harness series

Groundedness, citation accuracy, hallucination rate, abstention rate, regression pass rate and golden-set accuracy are all present in the RAG and Release dashboards. They are generated as a synthetic daily series, independent of the raw workflow/LLM/tool/retrieval spans — deliberately, because that mirrors exactly how a real implementation works: evals run on a schedule against sampled traffic, not inline on every request.

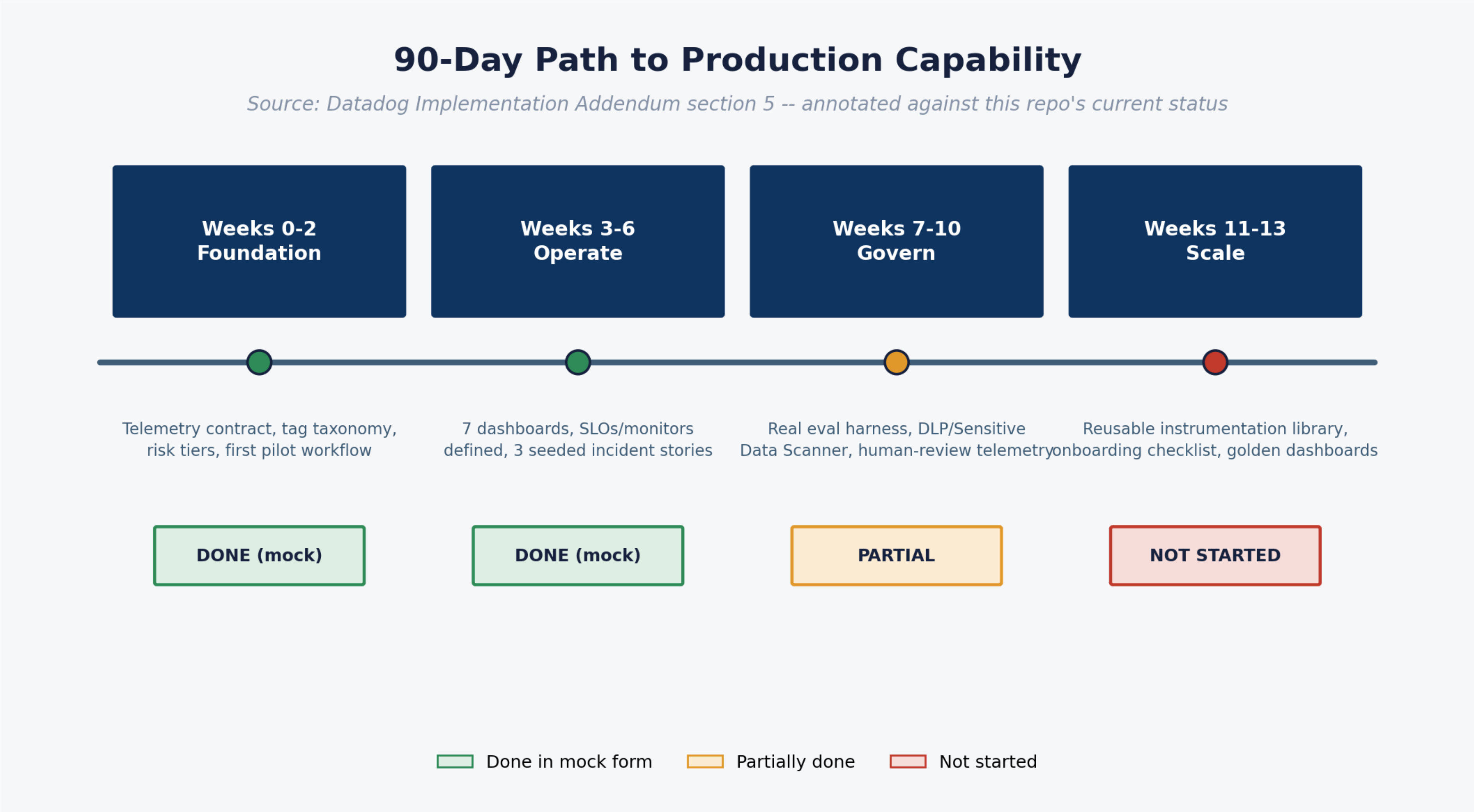
A real eval harness (LLM-as-judge, golden sets, human review sampling) is still required to produce these numbers for real — scoped into Weeks 7–10 of the roadmap. Presenting this caveat up front builds credibility with a technical audience; burying it costs credibility later.

04

Roadmap & next steps

The 90-day path from this mock demo to a live, governed pilot

Foundation → Operate → Govern → Scale



What it takes to go live

ORGANISATIONAL

- Named owner per accountability layer (Platform Eng, Security, Data Governance, Responsible AI, Product, Architecture)
- Risk tier defined for the pilot workflow
- Redaction/retention policy signed off before real content is logged
- Executive sponsor for the cost-avoidance / ROI narrative

TECHNICAL

- Datadog org with LLM Observability + APM entitlements
- SDK vs. OpenTelemetry instrumentation decision
- Approved model/tool registry for the pilot workflow
- Tag taxonomy agreed before scaling past the pilot
- On-call/alert-routing target to wire monitors and SLOs to

Sources: AI Observability Metrics Catalogue v1.0 • AI Observability — Datadog Implementation Addendum • OpenTelemetry GenAI semantic conventions • NIST AI RMF 1.0 • OWASP Top 10 for LLM Apps • Datadog LLM Observability docs